



DELHI PUBLIC SCHOOL NEWTOWN
SESSION: 2020-2021
FINAL TERM- ONLINE(POLLING) EXAMINATION

CLASS: IX
SUBJECT: PHYSICS

FULL MARKS: 30
TIME: 30 MINUTES

POLLING 1

1. The restoring force exerted by a spring (1)
 - acts opposite to the displacement of its free end
 - is a non contact force
 - is independent of magnitude of displacement of free end
 - cancels out the applied force on free end
2. It takes 0.2s for the pendulum bob to move from mean position to one end. The time period of the pendulum is (1)
 - 0.4s
 - 1.2s
 - 0.6s
 - 0.8s
3. The length of a simple pendulum is made one-fourth, then its time period becomes (1)
 - Four times
 - One-fourth
 - half
 - double
4. A solid of density 4000 kg/m^3 weighs 0.4kgf in air. It is completely inside water of density 1000 kg/m^3 . It's weight in water will be (2)
 - 0.1kgf
 - 0.16kgf
 - 0.3kgf
 - 0.5kgf
5. The R.D. of a body of weight 1.5 kgf is 3. It weighs 0.9kgf inside a liquid. The density of the liquid is (2)
 - 2700 kg/m^3
 - 900 kg/m^3
 - 1800 kg/m^3
 - 1200 kg/m^3

6. A galvanometer is a device to (1)
- Measure weak current
 - Measure strong and weak current
 - Connect parallel across the circuit
 - None of the above
7. Which of the following is not a social initiative for efficient use of energy resources? (1)
- Eco club activities
 - Imposing tax on sophisticated technologies
 - Collaboration with NGO's
 - Spreading awareness through mass-media
8. A compass needle placed at neutral point (1)
- Points to Geographic N-S direction
 - Will be resting with its axis parallel to N-S pole
 - Comes to rest in any arbitrary direction
 - Won't come to rest as net magnetic field is zero

POLLING 2

1. If the separation between two masses is reduced to half, the magnitude of gravitational force (1)
- Remain unaltered
 - Becomes twice its previous value
 - Is four times the previous value
 - Reduces to four times the previous value
2. If a force F acts on a 2kg mass initially at rest for 3s producing a velocity 15m/s, the value of F is (2)
- 10N
 - 22.5N
 - 15N
 - 11.25N
3. When a pendulum clock is inside a mine (1)
- Oscillations are faster
 - Oscillations increase first then decrease
 - Clock goes slow
 - Time period is unaltered
4. A solid weighs 32gf in air and 28.8gf in water. The volume of the body is (1)
- 3.02cm^3
 - 3.2cm^3
 - 2.88 cm^3

- 1.56cm^3
5. The distance between wave crest and its adjoining trough is (1)
 - Wavelength of the wave
 - Wave velocity
 - Half the wavelength of the wave
 - Half the wave velocity
 6. A charge 9.6C flowing through a conductor in 3s has is (2)
 - 6×10^{19} electrons associated with it
 - 6×10^{18} electrons associated with it
 - 3×10^{19} electrons associated with it
 - 2×10^{19} electrons associated with it
 7. The wire used as a standard resistor is generally made up of (1)
 - copper
 - manganin
 - lead-tin alloy
 - none of the above
 8. Which one of the following statements is incorrect? (1)
 - A hot body has more internal energy than an identical cold body
 - Liquids expand more than solids
 - Temperature alone indicates the quantity of heat energy contained in a body
 - Two bodies with same quantity of heat can differ in their temperatures

POLLING 3

1. The derived unit of work is (1)
 - $\text{Kgm}^2\text{s}^{-2}$
 - $\text{Kgm}^{-1}\text{s}^{-2}$
 - Kgms^{-2}
 - $\text{Kgm}^2\text{s}^{-1}$
2. When a corridor train suddenly starts sliding doors of some compartments may open, the reason is
 - the frame and the door both pick up the direction of motion of the train due to inertia
 - the equilibrium gets disturbed for the door frame is not in contact with the floor
 - sudden motion disrupts shock absorbers and cause motion in the door
 - The frame in contact with floor moves ahead, door slides in opposite direction due to inertia (1)

3. The pressure P_1 at a certain depth in river water and P_2 at the same depth in sea water are related as (1)
 - $P_1 = P_2$ as pressure at same horizontal level is same
 - $P_1 < P_2$
 - $P_2 < P_1$
 - None of the above
4. A solid weighs 30 gf in air and 24 gf inside a liquid of R.D. 0.6. The volume of the body is (2)
 - 6cc
 - 4cc
 - 5cc
 - 10cc
5. The line joining the point of incidence to the centre of curvature of a mirror (1)
 - is normal to the principal axis of the mirror
 - is normal to the tangent drawn on the spherical surface at the point of incidence
 - lie perpendicular to the focal plane
 - is double the focal length of the mirror
6. If speed of sound at 0°C be 330m/s, then at 20°C the speed will be (1)
 - 342.2m/s
 - 342m/s
 - 340m/s
 - 345.25m/s
7. When it is required for current to pass momentarily through the circuit, it is advisable to use (1)
 - rheostat
 - galvanometer
 - simple voltaic cell
 - tapping key
8. For a body floating half submerged in a liquid (1)
 - The centre of buoyancy will coincide with centre of gravity
 - Apparent weight of the body won't be zero
 - Centre of gravity of the displaced liquid lies below the centre of gravity
 - It's difficult to locate centre of buoyancy as no weight loss occur
9. Which of the following statements is true about magnetic lines of force? (1)
 - Magnetic lines of force are always closed.
 - Magnetic lines of force always intersect each other.
 - Magnetic lines of force tend to crowd far away from the poles of the magnet
 - Magnetic lines of force do not pass through the vacuum.